

Airline Dashboard

Executive Overview

Monthly airline passenger demand trends and key performance indicators

40K

Total Passengers

280.30

Avg Passengers

286%

Growth %

622

Peak Demand

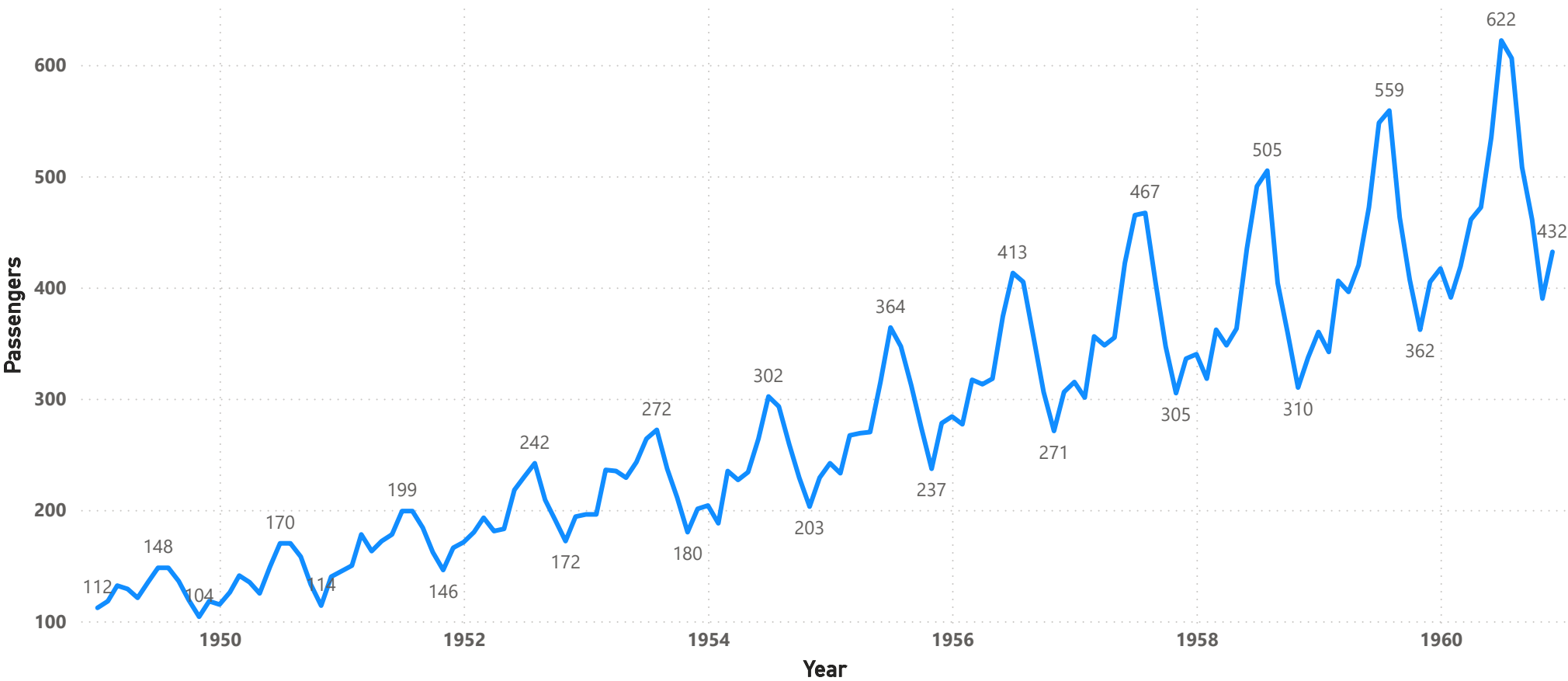
Executive Overview

Trend & Seasonality

Forecast vs Actual

Business Insights

Airline Passenger Demand Trend (1949-1960)



Insights: Passenger demand shows **a strong long-term upward trend with clear seasonal fluctuations**, indicating sustained growth in air travel and predictable yearly demand cycles.

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Forecast vs Actual

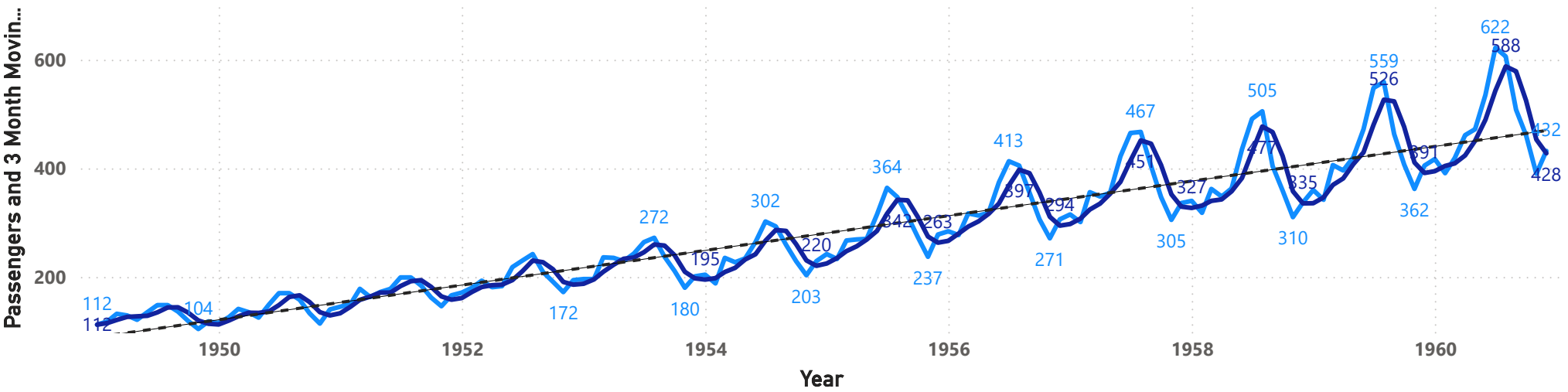
Business Insights

Trend & Seasonality Analysis

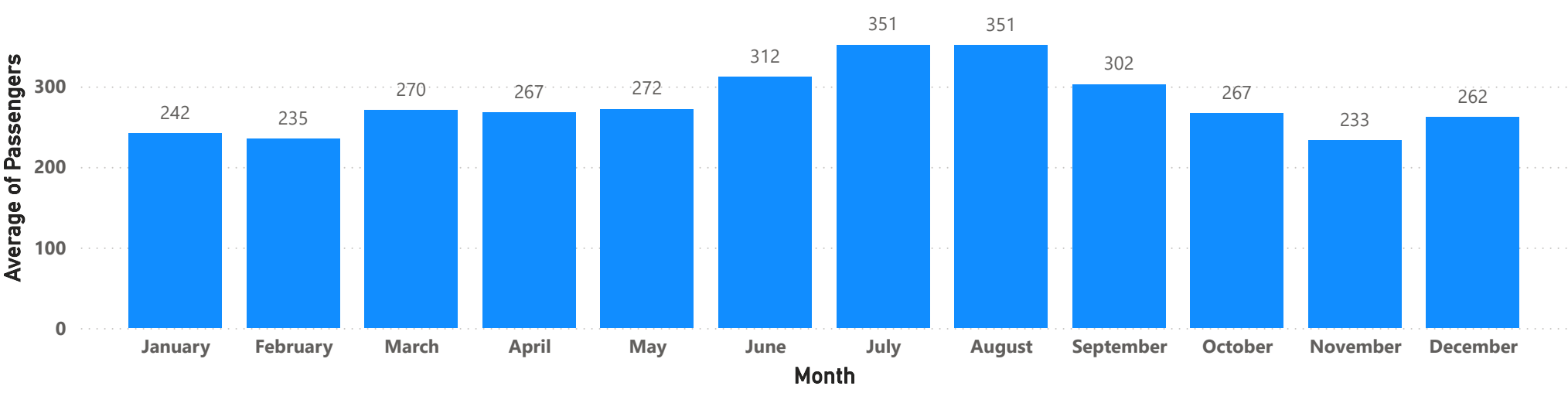
Monthly passenger patterns and seasonal demand behavior

Passenger Demand Trend Over Time

Passengers 3 Month Moving AVG



Monthly Seasonality Pattern



Insights: Passenger demand shows **strong annual seasonality with consistent yearly patterns. Peak travel occurs during June - August**, while **January - February consistently show lower demand**. These predictable seasonal cycles support improved capacity planning, scheduling optimization and resource allocation for airlines.

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Forecast vs Actual

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Forecast vs Actual (SARIMA Model)

Model performance and future passengers demand predictions

20.81

RMSE

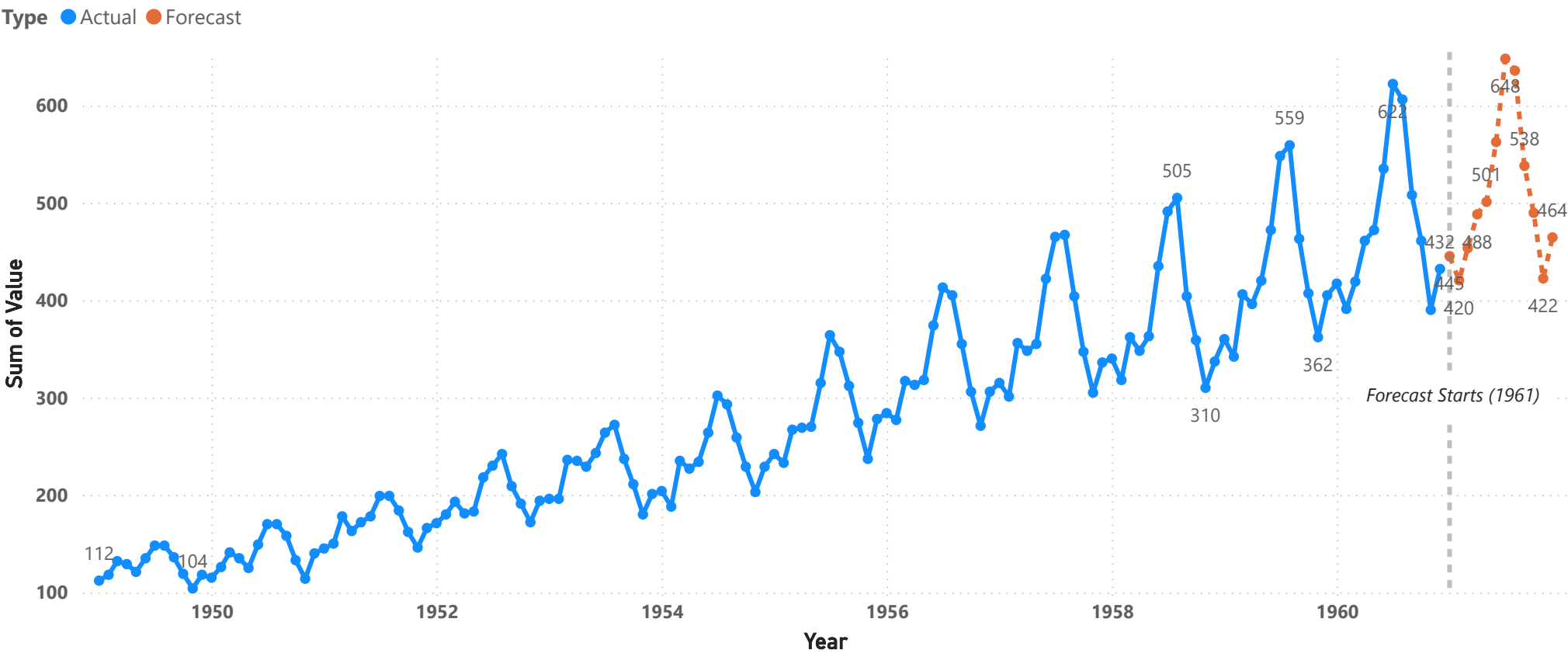
15.99

MAE

SARIMA(1,1,0)(1,1,0,12)

Model

Passenger Demand Forecast (Actual vs SARIMA Prediction)



Insights: The SARIMA model **successfully captures both long-term growth and strong seasonal patterns** in airline passenger demand. Forecast results continue the historical upward trend, indicating sustained growth in air travel demand. **Low RMSE (20.81) and MAE (15.99)** demonstrate good model accuracy and reliable predictive performance for future demand estimation.

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Business Insights

Actionable insights and recommendations for airline planning and operations

Executive Summary

This project analyzes historical airline passenger demand to identify long-term trends, seasonal patterns, and forecast future demand using SARIMA modeling. The insights support capacity planning and operational decision-making.

💡 Key Insights

- Airline passenger demand shows strong long-term upward growth with consistent and predictable yearly seasonality
- Peak demand occurs during June-August, while early-year months consistently show lower demand
- Growth trend indicates sustained expansion in global air travel demand over time

📊 Business Impact

- Improves airline capacity planning during peak travel seasons
- Improves airline scheduling, staffing and fleet allocation decisions
- Improves revenue forecasting and resource optimization
- Reduces risk of underutilization or overbooking capacity during seasonal spikes
- Enables proactive planning for predictable seasonal demand spikes

⚠️ Limitations

- Dataset is historical (1949-1960) and relatively small
- No external variables such as fuel price, economy, holidays included
- Assumes historical patterns continue into the future
- Model is based on univariate forecasting only

🚀 Next Steps

- Extend forecasting using ARIMAX with external variables
- Compare performance with Prophet or LSTM models
- Perform hyperparameter tuning to improve model accuracy
- Incorporate more recent airline datasets for better generalization